Flajolet martin:

```
data = [10, 20, 10, 30, 40, 20, 50, 60]
max_zeros = 0

for x in set(data):
    h = x % 11
    b = bin(h)[2:]
    z = len(b) - len(b.rstrip('0'))
    max_zeros = max(max_zeros, z)
    print(f"x={x}, Hash={h}, Binary={b}, Trailing zeros={z}")

print("Estimated distinct elements:", 2 ** max_zeros)
print("Actual distinct elements:", len(set(data)))
```

DGIM:

```
from collections import deque

stream = [1, 0, 1, 1, 0, 1, 0, 1, 1, 1]
N = 5
buckets = deque()

for t, bit in enumerate(stream, 1):
    if bit:
        buckets.appendleft([1, t])

        size = 1
        while sum(b[0] == size for b in buckets) > 2:
            i = [b[0] for b in buckets].index(size)
            buckets[i + 1][0] *= 2
            buckets.remove(buckets[i])

            size *= 2

    while buckets and buckets[-1][1] <= t - N:
        buckets.pop()

    print(f"Time {t}: Buckets = {list(buckets)}")

count = sum(b[0] for b in list(buckets)[:-1]) + (
    buckets[-1][0] // 2 if buckets else 0
)
print("Estimated number of 1s in last", N, "bits:", count)
```

Mapreduce:

```
# Read input file
with open("input.txt", "r") as f:
    text = f.read()

# Map phase
mapped = [(word, 1) for word in text.split()]
print("Mapper Output:", mapped)

# Reduce phase
count = {}
for word, value in mapped:
    count[word] = count.get(word, 0) + value

print("Word Count:", count)
```

Visualize in R:

Name,Marks,Hours,Department

A,80,5,IT

B,70,4,CS

C,90,7,IT

D,60,3,CS

E,85,6,IT

```
# Read CSV file
data <- read.csv("data.csv")

# 1. Boxplot
boxplot(data$Marks, main="Boxplot of Marks", col="skyblue")

# 2. Scatterplot
plot(data$Hours, data$Marks, main="Scatterplot",
    xlab="Study Hours", ylab="Marks", col="red", pch=19)

# 3. Histogram
hist(data$Marks, main="Histogram of Marks", col="lightgreen")

# 4. Pie Chart
pie(table(data$Department), main="Department Distribution",
   col=rainbow(2))

# 5. Bar Chart
barplot(table(data$Department), main="Department Count",
      col="orange", xlab="Department", ylab="Count")
```

implement Linear Regression, Logistic Regression, and Multiple Regression with plots, using a CSV file:

```
Hours,Marks,Pass,Attendance
2,40,0,60
3,50,0,65
4,55,0,70
5,65,1,75
6,75,1,80
7,85,1,90
8,95,1,95
```

```
# Read CSV file
data <- read.csv("data.csv")

# 1. Linear Regression
m1 <- lm(Marks ~ Hours, data=data)
plot(data$Hours, data$Marks, main="Linear Regression",
    xlab="Hours", ylab="Marks", pch=19, col="blue")
abline(m1, col="red", lwd=2)

# 2. Logistic Regression
m2 <- glm(Pass ~ Hours, data=data, family=binomial)
plot(data$Hours, data$Pass, main="Logistic Regression",
    xlab="Hours", ylab="Pass (0/1)", pch=19, col="blue")
x <- seq(min(data$Hours), max(data$Hours), length=100)
lines(x, predict(m2, newdata=data.frame(Hours=x),
```

```
        type="response"), col="red", lwd=2)

# 3. Multiple Regression
m3 <- lm(Marks ~ Hours + Attendance, data=data)
plot(data$Marks, predict(m3), main="Multiple Regression",
   xlab="Actual Marks", ylab="Predicted Marks",
   pch=19, col="blue")
abline(0, 1, col="red", lwd=2)

# Display results
summary(m1)
summary(m2)
summary(m3)
```

```
# Install and load igraph
install.packages("igraph")
library(igraph)

# Create a graph
edges <- c("A","B", "A","C", "B","C",
        "D","E", "D","F", "E","F",
        "C","D")

g <- graph(edges, directed = FALSE)

# Girvan-Newman algorithm
communities <- cluster_edge_betweenness(g)

# Display communities
print(communities)
cat("Number of communities:", length(communities), "\n")
print(membership(communities))

# Plot graph with communities
plot(communities, g,
    main = "Girvan-Newman Community Detection")
```